

Assignment 1: Market Risk (Volatility, Value-At-Risk and Expected Shortfall)

Due: Friday, July 10, 2026 @23:59

Objective: measure the market risk of a portfolio of your three stocks (covering volatility, correlation, beta, Value-at-Risk and Expected Shortfall) and analyze the results critically.

Required tasks:

- 1) Fetch three years of daily prices for your three stocks and your bloc's benchmark index. Compute returns, then inspect and handle any data-quality issues such as missing days and outliers, documenting what you cleaned.
- 2) Report each stock's annualized volatility and the correlation matrix, and comment on how much diversification your three names actually provide.
- 3) Estimate each stock's beta to your bloc's index (S&P 500 for North America, STOXX Europe 600 for Europe) and discuss systematic versus idiosyncratic risk. Test your own returns for the empirical regularities that any market-risk model must contend with - heavy tails (excess kurtosis) and volatility clustering.
- 4) Implement the 1-day 99% VaR by three methods (parametric variance-covariance, historical simulation, and Monte Carlo). Report how far the methods disagree for your portfolio and use that spread to argue how objective a single reported VaR number really is.
- 5) Compute Expected Shortfall alongside each VaR. Discuss what ES tells you about the tail beyond VaR, and test whether VaR is ever super-additive across your three names (so that the portfolio VaR exceeds the sum of the individual VaRs), which would show VaR failing to reward diversification.
- 6) Time-varying volatility: fit a GARCH(1,1) to one stock's returns. Report the parameters and the volatility persistence (alpha plus beta), produce a one-day-ahead conditional-volatility forecast, and compute a GARCH-based VaR. Contrast it with your constant-volatility VaR.
- 7) Backtest one VaR estimate over a rolling window: count the exceptions against the number expected at 99%, place the result in the Basel traffic-light zones, and interpret it.
- 8) Discuss assumptions, strengths, limitations, and real-world implications for your companies, framing it with the broader role of risk management.

Deliverable: Jupyter notebook: code, visualizations, narrative. $\leq 10,000$ chars.